Conjoint Cross-Attention Modeling and Joint Feature Calibrating for Remote Sensing Image Change Detection via a Triple-Double Network

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***Abstract*— Remote sensing (RS) image change detection (CD) based on deep learning (DL), has received increasing attention recently. However, the general independent learning of bi-temporal images ignores the relationship between them, falling short in learning the change information. In this article, a triple-double (TD) framework with the ability of conjoint cross-attention (CCA) modeling and joint feature calibrating is proposed for CD. Specifically, the TD framework composed of a triple-branch encoder and double-branch decoder is constructed to extract diverse features and acquire changed maps with the guidance of original edge cues. To enhance the perception of the connection between the bi-temporal features, the multiscale difference guidance (MDG) module and CCA module are designed for the dual-branch encoder, wherein the CCA introduces a novel and efficient rule for modeling the affinity in spatial and channel dimensions simultaneously. Furthermore, a joint feature calibration (JFC) module is introduced to enhance the expression of feature diversity in the joint features within the single-branch encoder. Experimental results on three public datasets demonstrate the superiority of the proposed method compared to the state-of-the-art (SOTA) methods.**

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***Index Terms*— Change detection (CD), conjoint cross-attention (CCA) module, joint feature calibration (JFC), remote sensing (RS), triple-double (TD) framework.**

I. Introduction

CD TASK is a process of analyzing and identifying the areas of change between image pairs in the same scene at different times [1]. With the development of RS technology, CD tasks can obtain a broader view of landscapes, possessing the ability to efficiently monitor and survey larger-scale information on the surface [2]. RS CD has been extensively applied in many fields, such as environmental monitoring [3], natural damage assessment [4], urban planning and extension estimation [5], and so on. In such a context, striving to obtain high-precision change maps for CD tasks becomes increasingly crucial for protecting Earth resources.

Faced with these issues, researchers have proposed various methods continuously, which can be roughly classified into traditional methods and DL-based methods. Before the prevalence of DL technology, CD methods are mainly occupied by traditional methods. Early traditional methods are based on pixel-level analysis such as image differencing [6], image quantification [7], and change vector analysis (CVA) [8], which generate difference images through implementing certain algebraic-based operations on RS images and then obtains binary CD maps by thresholding or cluster analysis. The other CD techniques based on machine learning methods such as support vector machines [9], decision trees (DTs) [10], and random forest regression [11] are also flourishing, which extract manual features from bi-temporal images and analyze them to identify the changed areas.

Compared to traditional methods, DL-based methods mine the spatial–temporal features on bi-temporal images via the great learning and modeling ability of neural network (e.g., convolutional neural network (CNN) [12] and recurrent neural network (RNN) [13]) rather than manually feature extracting. Early, most DL-based CD methods used CNN to mine bi-temporal image information due to its powerful feature extraction capability. Next fully convolutional network (FCN) [14] is also employed in the CD task for dense prediction. The majority of FCN-based methods adopt an encoder–decoder architecture, which can be further divided into "early fusion" methods and "late fusion" methods. The "early fusion" methods treat the concatenations or differences of image pairs as the

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network inputs, typically applied in single-branch networks. In contrast, the "late fusion" methods utilize the individual bi-temporal images as the network input, commonly employing a dual-branch (Siamese) architecture to independently learn feature information from each image. The single-branch network only works for the combined bi-temporal images to extract joint features while ignoring the features from individual images. As for the dual-branch network, though the separate images are fed into Siamese feature extractors to extract features, respectively, the relationship and change information of the bi-temporal inputs are not perceived sufficiently.

Recently, researchers have introduced various attention-based methods for CD to improve the performance of neural networks [15], [16], such as convolutional block attention module (CBAM) [17]. To capture long-range dependencies, nonlocal block [18] is also introduced into the CD networks [19], [20], [21]. The utilization of the various attention mechanisms has indeed improved the CD prediction accuracy. However, the issue of attention loss and bias suppresses further improvements in performance. To be specific, the interchannel relationship can be well modeled in channel attention (CA) while neglecting the relevance of the spatial information. Similar bias occurs in spatial attention (SA), with sufficient modeling of spatial affinities yet scarce concentration on the channel dependency. To alleviate the attention bias, [19] and [22] implement the channel and SA in a sequential and parallel manner, respectively, resulting in the problem of higher memory and computation costs. In addition, no matter whether deploying the channel and SA in a sequential or parallel manner, the integrity and interaction of the context between channel and spatial dimensions are left out.

To address the issues mentioned above, this article proposes a TD framework with CCA modeling and joint feature calibrating. To mine the potential joint feature representation, an extra single-branch encoder is linked to the dual-branch encoder, making up the triple-branch encoder. Specifically, the multistage enhanced difference features and joint features from the triple-branch encoder are fused and sucked into the double-branch decoder for analyzing and recognizing the main change regions and the auxiliary edge structure. The main contributions of this article are summarized as follows.

1) A TD framework is constructed to concurrently learn the original joint features and difference features, utilizing edge structure cues to supervise the CD.
2) For the dual-branch encoder, we design an MDG module for leveraging the multiscale enhanced difference information to guide the bi-temporal features. Moreover, a CCA module, employing a novel attention mechanism, is introduced for the refined bi-temporal features from the MDG module to capture spatial–temporal relationships for pushing the model to focus more on changed areas.
3) A JFC module is introduced and incorporated for constructing the single-branch encoder, facilitating the learning and calibration of the original joint features.

The rest of this article is structured as follows. In Section II, a brief reviewing of related works is given. Section III describes the specifics of the proposed network. The comprehensive experimental results and ablation studies on three CD datasets are conducted in Section IV. The final Section V draws a conclusion of this article.

For the convenience of readers, we collect the key acronyms and notations employed throughout the article in a given table, as shown in Nomenclature.

NOMENCLATURE

Acronym	Description
RS	Remote sensing.
CD	Change detection.
DL	Deep learning.
TD	Triple-double.
MDG	Multiscale difference guidance.
CCA	Conjoint cross-attention.
JFC	Joint feature calibration.
CBR	Convolution and batchnorm and Relu.
BSA	Bi-dimensional spatial aggregation.
SA	Spatial self-attention.
CA	Channel self-attention.
Notation	**Description**
C	Input feature dimension.
H	Height of the input feature map.
W	Width of the input feature map.
S	Substitute to H and W.
T	Bi-temporal input images.
F	Bi-temporal input features.
O	Bi-temporal output features.
D	Difference features.
D_s	Aggregating difference features.
D'	Weight difference features.
P_h	Horizontal spatial priors.
P_v	Vertical spatial priors.
P_s	Global spatial priors.
S_w	Softmax function in width-dimension.
S_h	Softmax function in height-dimension.
F_h	Horizontal reconstructed features.
F_v	Vertical reconstructed features.
A	Cross-temporal affinity matrices.
F_c	Multiscale calibration vector.
α	Learnable coefficients.
β	Learnable coefficients.
r	Dilation rates.
σ	Sigmoid function.
γ	Tradeoff weight.

II. RELATED WORKS

A. Traditional CD Methods

In the early phase of the investigation of CD, the researches mainly focus on unsupervised methods to achieve the CD task. Traditional unsupervised methods typically involve comparing data from different periods and looking for their differences. One commonly used traditional method is the image differencing technique [6], which involves subtracting the pixel values of two images captured at different times and analyzing the resulting difference image. Another method is the image rationing technique, which involves dividing the pixel values

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of two images captured at different times. This can help highlight changes in vegetation, land cover, or other features. In addition, there are methods such as CVA [8] and principal component analysis (PCA) [23] that are widely used for CD. However, these methods mainly rely on the extraction of the spectral information while ignoring the role of the spatial relationship.

Machine learning-based methods aim to integrate both spectral and spatial information for analyzing the difference map of the bi-temporal images. For example, Moser et al. [24] applied the Markov random field (MRF) to integrate spectral features and spatial context. In order to improve the expression and recognition of features, DTs [10], random forest regression [11], and Bayesian classification [25] are also increasingly proposed for CD. However, due to the reliance on empirical design and the poor robustness of CD, these traditional methods cannot perform well in complicated scenes.

B. DL-Based CD Methods

With the development of DL architectures, some DL-based CD methods are proposed, which automatically learn and extract features from bi-temporal RS images for change prediction. For instance, Liu et al. [26] proposed a network with two parallel CNN channels for CD from bi-temporal SAR images. Later, FCN and its variants (e.g., UNet) have also been introduced for RS CD task [27], [28]. Zhuang et al. [29] introduced an enhanced generalized likelihood ratio test (GLRT) [30] technology that can dynamically choose the optimal window for each pixel in order to obtain the difference images between the bi-temporal synthetic aperture radar (SAR) images. However, these "early fusion" methods ignore the original information of individual images, resulting in insufficient learning of bi-temporal images.

Siamese networks usually use two replicated encoders with shared weights to separately extract the bi-temporal features and fuse them to obtain the changed feature map. Daudt et al. [31] first introduced a Siamese network for CD, where one early fusion network (FC-EF) and two fully convolutional Siamese networks (FC-Siam-conc and FC-Siam-diff) are proposed to explore the impact of training methods and fusion strategy on bi-temporal image inputs. Chen et al. [32] designed a multiscale convolution unit for the network to perceive the objects at various scales. Yang et al. [33] proposed an asymmetric Siamese network to locate and identify the changes at the semantic level. In order to obtain more distinguishable feature pairs, Liu et al. [16] designed a stacked attention module consisting of several CBAMs [17] to enhance the multiscale features from the Siamese encoder. The employment of deep supervision in [15] is introduced to assist the intermediate layers in capturing more meaningful features. However, this separate feature extraction manner ignores the original intercorrelation between the bi-temporal images.

To enhance the changed information representations and suppress the interference of the pseudo-changed one, the attention mechanism emerges and becomes increasingly popular in the field of CD. Recently, researchers have utilized CA, SA, and self-attention mechanisms synthetically to further improve the performance [17], [34], [35]. The self-attention mechanism has the ability to capture long-range dependence of features. Similarly, the contextual affinity modeling of self-attention can be divided into channel and spatial dimensions. Chen and Shi [19] and Chen et al. [22] combined the channel and SA mechanism in a sequential and parallel manner, respectively, to capture the global spatial–temporal relationships for more discriminative features. However, these simple combinations of two attention mechanisms ignore the intrinsic spatial-channel interactions and cause the computational burden, which urges the researches to find an effective and lightweight manner for modeling the channel and spatial contextual affinity synergistically.

III. Methodology

The overall architecture of the proposed network is illustrated in Fig. 1. The bi-temporal inputs are fed into the triple-branch encoder to extract difference features and joint features simultaneously. For enhancing the discriminative features representation and mutual information learning, the MDG module and CCA module are designed in a sequential manner within the dual-branch encoder. As for joint feature learning, JFC modules are introduced to construct the single-branch encoder to explore the inner correlation of image pairs. Finally, the features from the triple-branch encoder are sucked into the double-branch decoder to integrate the composite features and generate the changed map.

A. TD Framework

Considering the indispensability of the original joint features and difference features, a specific triple-branch encoder, consisting of the dual-branch encoder and single-branch encoder, is employed to extract both two types of features simultaneously and enhance the capability of detecting changes in the input images. Besides, a double-branch decoder composed of a change decoder and an edge decoder is introduced in the network, which accomplishes feature fusion for change prediction and leverages the original edge detail information of joint features for auxiliary supervision. It helps extract comprehensive and discriminative representations, while the guidance process facilitates more accurate and precise CD results.

As depicted in Fig. 1, the bi-temporal RS images T_1 and T_2 are separately fed into the dual-branch encoder, where the basic block is composed of three combination operations: a convolution layer, a BatchNorm layer, and a ReLU layer (abbreviated as CBR). Then the extracted bi-temporal features are forwarded to the MDG and CCA modules for further information learning. Simultaneously, a single-branch encoder (constructed by JFC modules) is specifically designed for learning and calibrating the joint features, utilizing the concatenated image as its input. The features captured by the single-branch encoder provide not only change information for change prediction in the change decoder phase but also edge structure information for edge detection in the edge decoder phase. Overall, the triple-branch encoder facilitates the associative learning of individual and joint information.

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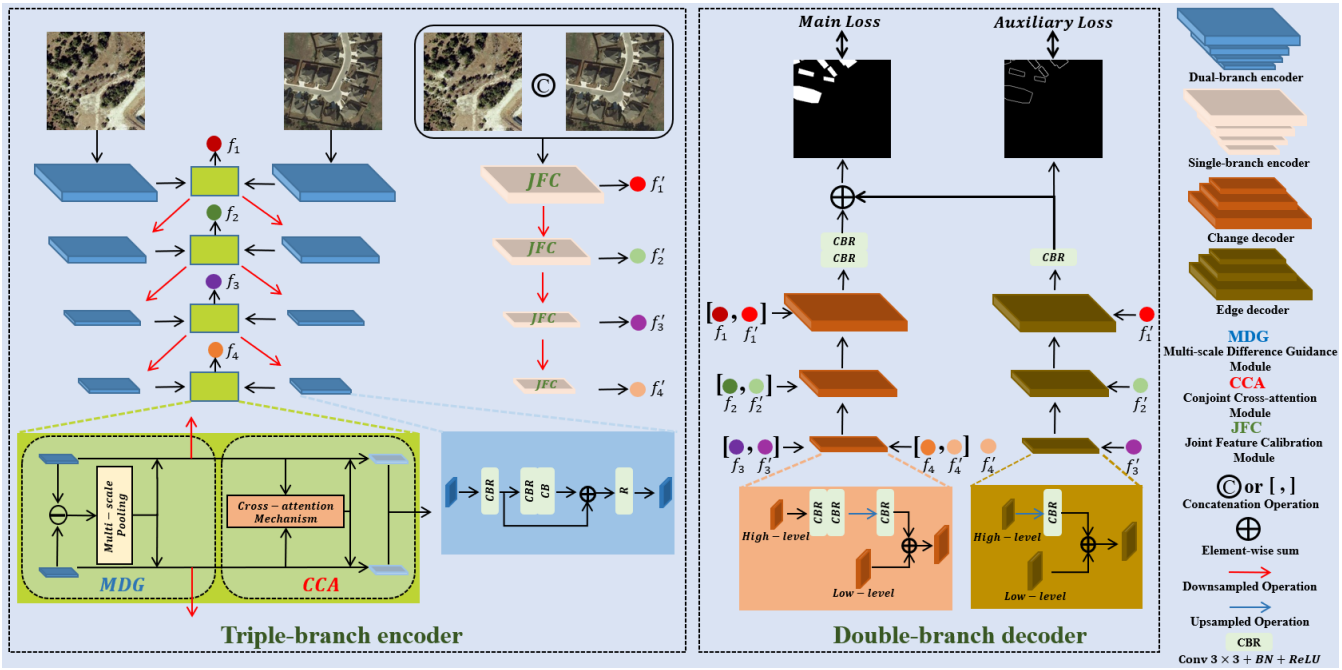


Fig. 1. Architecture of the proposed network.

Given the rich features of a triple-branch encoder, we design a double-branch decoder consisting of a change decoder and edge decoder to incorporate them and reconstruct the image for prediction, where CBR operations are also repeatedly implemented in each decoder. As illustrated in the double-branch decoder part in Fig. 1, the change decoder merges the information from dual-branch and single-branch encoders in different scales and generates the CD result supervised by the main loss. Unlike the change decoder, the inputs of the edge decoder exclusively come from the single-branch encoder, which leverages the fine-grained spatial details captured in lower-level feature maps. Moreover, contour labels are generated from the CD ground truth and utilized to train the edge branch by deep supervision.

B. MDG Module

To derive change information, relying solely on straightforward subtraction operations between bi-temporal features from the dual-branch encoder is insufficient and vulnerable to noise interference. Meanwhile, by drawing insights from the pyramid pooling module in PSPNet [36] and the unequal channel pyramid pooling module in SA-FFNet [37], an MDG module is designed to leverage the multiscale enhanced difference information for guiding the bi-temporal features in the dual-branch encoder, as illustrated in Fig. 2.

The bi-temporal features $\{F_1, F_2\} \in \mathbb{R}^{C\times H\times W}$, originating from each layer of the dual-branch encoder, serve as inputs to the MDG module, where H, W, and C represent the height, width, and channel dimension. Then, an operation involving the subtraction of F_1 and F_2 is executed to yield the initial difference information map $D \in \mathbb{R}^{C\times H\times W}$, as follows:

$$D = |F_1 - F_2| \tag{1}$$

where $|\cdot|$ represents the absolute value operation. Subsequently, D is subjected to a pyramid pooling procedure to acquire diverse difference information from multiscale feature maps. More precisely, the pyramid pooling procedure encompasses three distinct-scale average pooling operations ($\text{AvgPool}_s(\cdot)$ with the pooling size s), succeeded by a 1×1 convolutional operation ($\text{Conv}_{1\times1}$), as follows:

$$D_s = \text{Conv}_{1\times1}\big(\text{AvgPool}_s(D)\big). \tag{2}$$

Then three parallel difference features $D_1 \in \mathbb{R}^{C\times1\times1}$, $D_2 \in \mathbb{R}^{C/4\times2\times2}$, and $D_3 \in \mathbb{R}^{C/16\times4\times4}$, which condense both global information and local details across spatial and channel dimensions, are generated accordingly.

To consolidate the multiscale difference descriptors, we employ flattening and summation operations on three feature maps. Inspired by the CA operation applied in [38], we pass the fused descriptors through a multilayer perceptron MLP($\cdot$) to enhance the utilization of the aggregated difference information. Then, a sigmoid activation operation σ is utilized to generate a weight feature vector D' for calibrating the bi-temporal features in channel dimension, as shown in the following equation, where flatten($\cdot$) represent the flatten

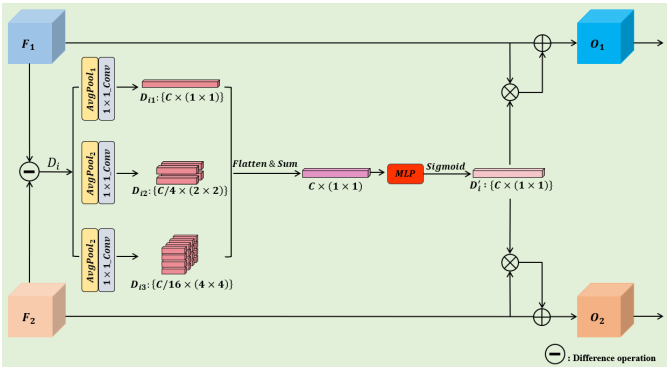


Fig. 2. Structure of the MDG module.

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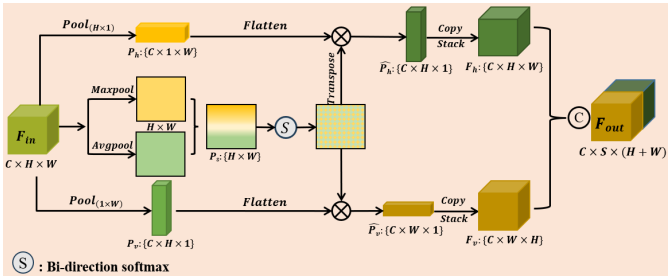


Fig. 3. Structure of the BSA block.

operation

$$D' = \sigma\left(\mathrm{MLP}\left(\mathrm{flatten}\left(D_{s(1)}\right) + \mathrm{flatten}\left(D_{s(2)}\right) + \mathrm{flatten}\left(D_{s(3)}\right)\right)\right), \quad s = [1, 2, 4]. \tag{3}$$

Finally, we perform sequential element-wise multiplication and summation operations between the input bi-temporal features $\{F_1, F_2\}$ and the enhanced difference descriptors D' to obtain the final guided bi-temporal outputs. The specific formula is defined as

$$O_1 = F_1 \odot D' + F_1 \tag{4}$$

$$O_2 = F_2 \odot D' + F_2 \tag{5}$$

where O_1 and O_2 represent the guided outputs of the module. The outputs are supplied as input to the corresponding CCA module for the subsequent procedures while being upsampled to serve as input for the next encoder stage. $\odot$ denotes the element-wise multiplication. Through multiplication operations, attention values can be broadcasted correspondingly.

Since the MDG module only involves some convolution, pooling, fully connected layers, and some simple calculation operations, its computational complexity is linearly related to the size of the feature map. Therefore, given the input feature with a size of $C \times H \times W$, the computational complexity of MDG is $O(\mathrm{CHW} + C^2)$.

With the guidance of multiscale difference information, the network can concentrate more on the really changed objects but less on the pseudo changes. The effectiveness of the MDG module has been proved in the ablation studies in Section IV.

C. CCA Module

To capture the interaction between bi-temporal features without a high computational burden, we propose a CCA module employing a novel and efficient rule for a nonlocal attention mechanism, which highlights relevant details and ignores the irrelevant or meaningless parts of features.

As preparation, a BSA block is introduced on the CA mechanism to synergistically model spatial and channel dependencies while minimizing computational overhead. The details of BSA are illustrated in Fig. 3.

In the BSA, the input $F_{\mathrm{in}} \in \mathbb{R}^{C\times H\times W}$ is initially performed by width-wise and height-wise global average pooling along two individual branches with the size of $H \times 1$ and $1 \times W$, respectively. The purpose of the bi-directional pooling operations is to compress spatial contextual information horizontally and vertically. Accordingly, the generated horizontal and vertical spatial descriptors of F_{in} are denotes as $P_h \in \mathbb{R}^{C\times 1\times W}$ and $P_v \in \mathbb{R}^{C\times H\times 1}$. Meanwhile, inspired by SA in [17], we aggregate the spatial information of F_{in} by stacking the average-pooled features and max-pooled features to generate the global spatial priors $P_s \in \mathbb{R}^{1\times H\times W}$, the process is denoted as

$$P_s = f_{7\times 7}\left(\left[\mathrm{AvgPool}(F_{\mathrm{in}}); \mathrm{MaxPool}(F_{\mathrm{in}})\right]\right) \tag{6}$$

where $f_{7\times 7}$ denotes a 7×7 convolution operation, $[\cdot;\cdot]$ denotes concatenation operation, and $\mathrm{AvgPool}(\cdot)$ and $\mathrm{MaxPool}(\cdot)$ represent the average-pooling and max-pooling operations, respectively. Then, we adopt a bi-direction softmax function for normalizing the 2-D map P_s in both H and W dimensions. The normalized row-wise and column-wise affinity matrices are multiplied by the spatial descriptors P_h and P_v, respectively, for calibration. In preparation for this calibration, P_h and P_v are flattened into the 2-D feature maps $P_h{}' \in \mathbb{R}^{C\times W}$ and $P_v{}' \in \mathbb{R}^{C\times H}$, which represent the calibrated horizontal and vertical spatial descriptors, respectively. The process is formulated as

$$\hat{P}_h = P_h{}' \otimes S_w(P)^{\mathrm{T}} \tag{7}$$

$$\hat{P}_v = P_v{}' \otimes S_h(P)^{\mathrm{T}} \tag{8}$$

where S_w and S_h are the softmax functions in width-dimension and height-dimension, respectively, $\cdot^{\mathrm{T}}$ denotes the matrix transpose operation, and $\otimes$ denotes the matrix multiplication operation. Then, we reshape $\hat{P}_h$ and $\hat{P}_v$ followed by copying and stacking operation to obtain the reconstructed features $\{F_h \in \mathbb{R}^{C\times H\times W}, F_v \in \mathbb{R}^{C\times W\times H}\}$, which are concatenated together as the output of BSA block, denoted as $F_{\mathrm{out}} \in \mathbb{R}^{C\times S\times(H+W)}$, where the coefficient S is equal to H and W numerically.

Based on the BSA, the pipeline of the CCA module is shown as Fig. 4. To facilitate affinity modeling and promote interaction between the enhanced bi-temporal features obtained from the MDG modules, we adopt a cross-manner approach with the nonlocal attention mechanism. The input bi-temporal features $\{F_1, F_2\} \in \mathbb{R}^{C\times H\times W}$ are first fed into three parallel BSA blocks to mine the spatial relationships and the outputs are denoted as six feature vectors: $\{Q_1, K_1, V_1, Q_2, K_2, V_2\} \in \mathbb{R}^{C\times S\times(H+W)}$. To find the feature interactions between F_1 and F_2, we conduct an element-wise multiplication operation between Q_1 and transposed K_2, as well as Q_2 and transposed K_1, followed by a softmax function to calculate the affinity. The expression is as follows:

$$A_1(i,j) = \frac{\exp\left(Q_1(i)\cdot K_2^{\mathrm{T}}(j)\right)}{\sum_{i=1}^{C}\exp\left(Q_1(i)\cdot K_2^{\mathrm{T}}(j)\right)} \tag{9}$$

$$A_2(i,j) = \frac{\exp\left(Q_2(i)\cdot K_1^{\mathrm{T}}(j)\right)}{\sum_{i=1}^{C}\exp\left(Q_2(i)\cdot K_1^{\mathrm{T}}(j)\right)} \tag{10}$$

where $\{A_1(i,j), A_2(i,j)\} \in \mathbb{R}^{(H+W)\times C\times C}$ represent the measurement of the correlation between the ith channel of Q_1 and the jth channel in K_2, the ith channel of Q_2 and the jth channel in K_1 at a specific spatial pixel position.

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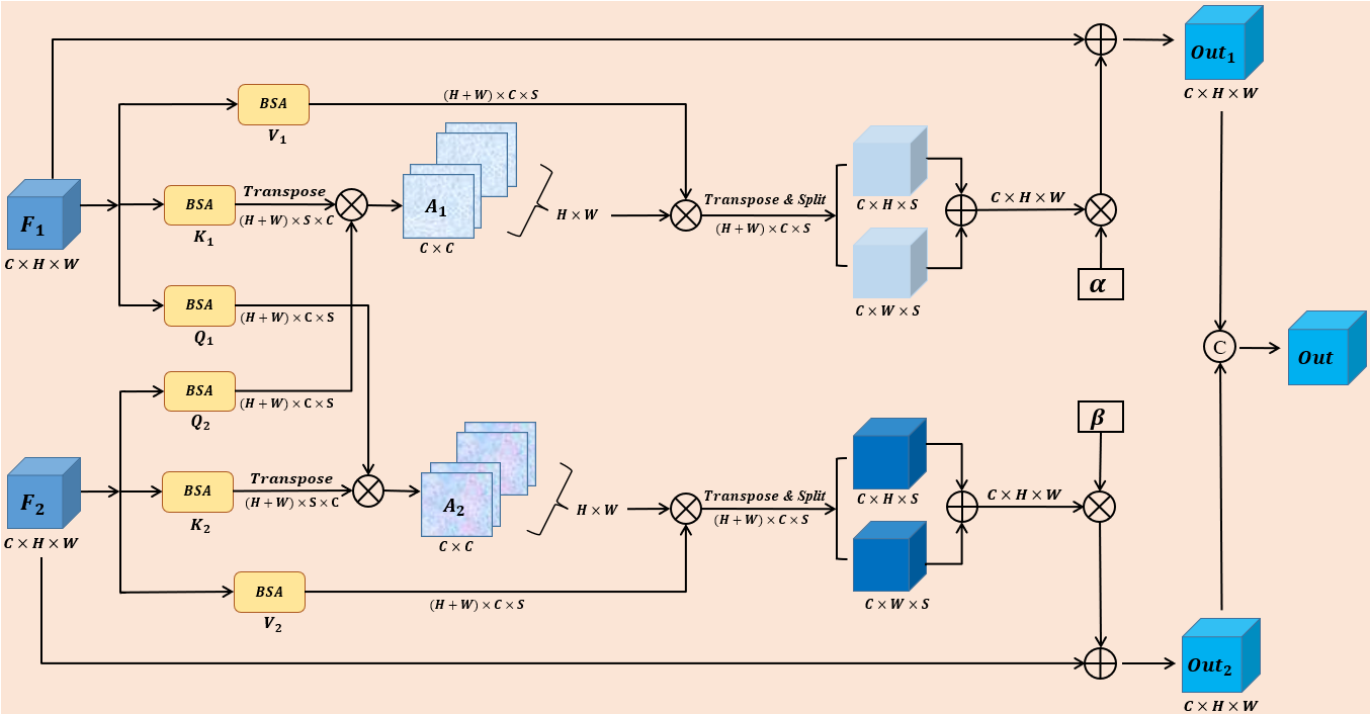


Fig. 4. Pipline of CCA module.

Then, we conduct another two matrix multiplications between A_1 and V_1, between A_2 and V_2 to calibrate channel responses. Subsequently, we split the results into two parts of sizes $\mathbb{R}^{C\times H\times S}$ and $\mathbb{R}^{C\times W\times S}$, respectively, and then conduct summation operation on them to further integrate spatial and channel contextual information. Finally, we utilize two learnable parameters to multiply the two integrated results.Afterward, we perform an element-wise summation with F_1 and F_2 to obtain the bi-temporal outputs, namely, $\{\text{Out}_1, \text{Out}_2\} \in \mathbb{R}^{C\times H\times W}$.

Finally, the output of CCA module Out is generated by concatenating Out_1 and Out_2. The above process can be expressed as

$$O_1(j) = \alpha \sum_{i=1}^{C} A_1(i, j) \cdot V_1(j) + F_1(j) \tag{11}$$

$$O_2(j) = \beta \sum_{i=1}^{C} A_2(i, j) \cdot V_2(j) + F_2(j) \tag{12}$$

$$O = [\text{Out}_1, \text{Out}_2] \tag{13}$$

where $\text{Out}_1(j)$ and $\text{Out}_2(j)$ represent the jth channel of Out_1 and Out_2, α and β are both learnable coefficients and Out represents not only the output of CCA module but also the output of the corresponding stage encoder.

Given an input feature with a size of $C \times H \times W$, the computational complexity of the SA mechanism is $O((HW)^2C)$ while the SA mechanism has a complexity of $O((C)^2HW)$. In order to model 3-D dependencies, sequential and parallel architectures separately calculate spatial and channel-wise correlations, resulting in higher complexity and demanding more GPU memory. The CCA module introduces a novel computational approach by jointly modeling spatial and channel dimensional correlations, with a computational complexity of $O(C^2(H + W) * S)$.

In summary, the proposed CCA module facilitates further interaction with the bi-temporal features through a conjoint manner of modeling the spatial and channel contextual information. Compared to the general CA mechanism, the CCA additionally takes into account spatial correlations in both the height dimension and width dimension while maintaining a similar order of computational complexity.

D. JFC Module

To enhance the diverse representation of the joint features, we design a JFC module to construct a single-branch encoder. The core of the block lies in generating a multiscale calibration vector and using it to calibrate the joint features. First, in order to perceive the multiscale objects, we employ atrous convolutions [39] with multiple atrous rates in a parallel manner to capture context information at various scales. Then, to refine the integrated features extracted from different receptive fields and emphasize the regions of interest, we employ an attention-based method for adaptive calibration of the joint features.

The overall structure of the JFC module is shown in Fig. 5. As a preliminary step in feature extraction with a fixed receptive field, we apply a 1×1 convolution operation on the inputs $F_{\text{in}} \in \mathbb{R}^{C\times H\times W}$. Then, we employ four parallel dilated 3×3 convolution layers with dilation rates $\{1, 2, 4, 6\}$ to extract the features from multiple receive fields, yielding four feature maps $\{F_1, F_2, F_3, F_4\} \in \mathbb{R}^{C/4\times H\times W}$ with one-quarter channel numbers and same spatial size of F_{in}. The process can be expressed as

$$F_i = \text{Conv}_{3\times 3}^{r_i}\left(\text{Conv}_{1\times 1}^{r=1}(F_{\text{in}})\right) \quad i = 1, 2, 3, 4 \tag{14}$$

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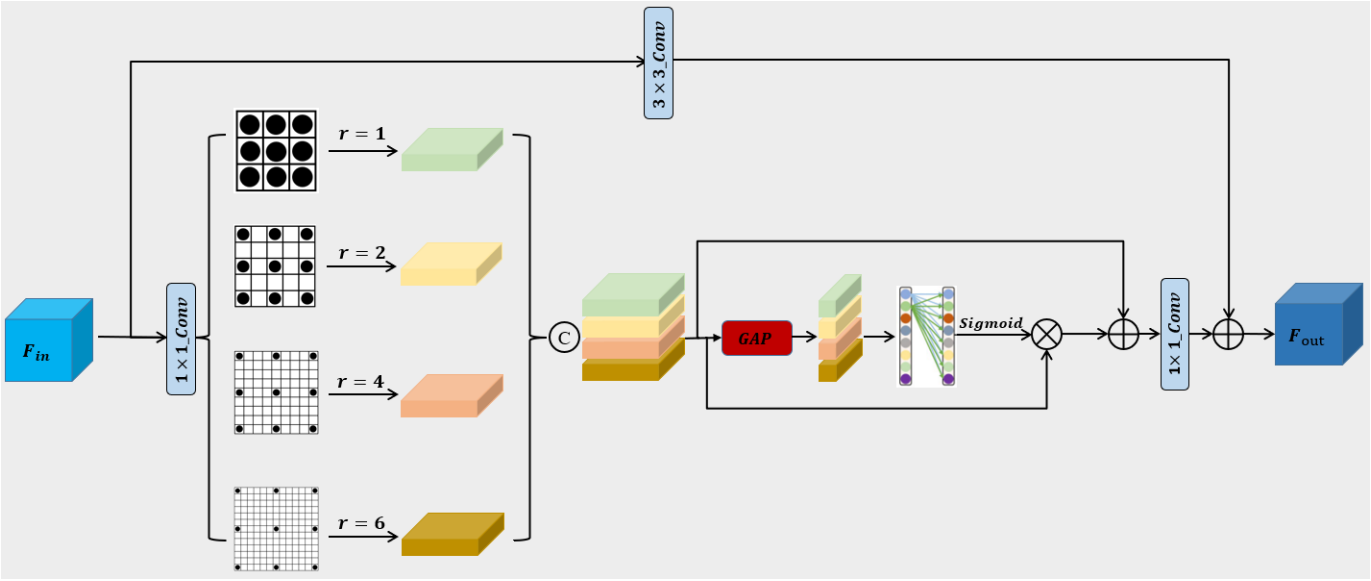


Fig. 5. Structure of JFC module.

where i denotes the ith atrous convolution, $\text{Conv}_{3\times3}^{r_i}$ represents the convlutional layer with the filter size of $k \times k$ and dilation rates $r_i = 2 \times (i-1)$ when $(i \neq 1)$ or 1 when $(i = 1)$. The integrated feature $F' \in \mathbb{R}^{C\times H\times W}$ is obtained by concatenating the four parallel feature maps in channel dimension. Following the previous steps, we perform a global average pooling operation on the integrated feature map to squeeze the global spatial information of each channel. Subsequently, we pass the squeezed feature map through two fully connected layers to enhance the interrelationship of feature representations across different channels. And then the multiscale calibration vector $F_c \in \mathbb{R}^{C\times1\times1}$ is generated by a sigmoid activation function. The above calculation process is as follows:

$$F' = \text{Concat}(F_1, F_2, F_3, F_4) \tag{15}$$

$$F_c = \sigma\left(fc_2\left(fc_1\left(\text{GlobalAvgPool}\left(F'\right)\right)\right)\right) \tag{16}$$

where Concat$(\cdot)$ denotes concatenation operation in channel dimension, GlobalAvgPool$(\cdot)$ stands for the global average pooling operation, and fc_1 and fc_2 represent two sequential fully connected layers. To adaptively emphasize the useful regions, we utilize the multiscale calibration vector F_c to calibrate the integrated feature map F' by a multiplication operation. Furthermore, we introduce a residual operation between the fused features and the calibrated features followed by a 3×3 convolution for the purpose of preserving the original information of integrated features. Finally, we add another residual branch consisting of a 3×3 convolution to establish a connection between the original input features and the refined features, which generates the final output $F_{\text{out}} \in \mathbb{R}^{C\times H\times W}$. The process is formulated as follows:

$$F_{\text{out}} = \text{Conv}_{3\times3}((F_c \odot F_{\text{fuse}} + F_{\text{fuse}})) + \text{Conv}_{3\times3}(F_{\text{in}}). \tag{17}$$

Since the types of operations applied in JFC are similar to those in MDG, the computational complexity of JFC is also $O(\text{CHW} + C^2)$, with the input size of $C \times H \times W$.

In the proposed JFC module, the incorporation of multiple atrous convolutions in parallel allows us to perceive objects at various scales. This enables accurate localization and improves the recognition of multiscale objects of interest. Furthermore, by employing a CA-based method, we can adaptively calibrate the features extracted from different receptive fields. The calibration process prioritizes features at meaningful scales, enhancing the effectiveness of the module in learning joint features during the single-branch encoder phase.

E. Loss Function

The double-branch decoder in the proposed network includes a change prediction branch and an edge auxiliary prediction branch. The edge probability map is used to compute the edge auxiliary loss in a deeply-supervised [40] manner. The change feature map from the change decoder and the edge feature map from the edge decoder are concatenated followed by a CBR and convolution operations to generate the refined change probability map.

In the training process, we select binary cross-entropy (BCE) loss as the main loss function and utilize it concurrently as the auxiliary loss function. The function can be formulated as

$$L_{\text{bce}} = -\frac{1}{N}\sum_{1}^{N}\left[y_i \log(p_i) + (1-y_i)\log(1-p_i)\right] \tag{18}$$

where N is the total number of pixels, $y_i \in \{[0, 1\}$ and $p_i \in [0, 1]$ denote the value of pixel i in change label map and predicted change map, respectively. The total training loss L_{total} is expressed as

$$L_{\text{total}} = L_{\text{bce}}(P_{\text{final}}, Y) + \lambda L_{\text{bce}}\left(P_{\text{edge}}, Y_{\text{edge}}\right) \tag{19}$$

where λ stands for the tradeoff weight, P_{final} and P_{edge} represent the final change probability map and edge probability map, respectively, and Y and Y_{edge} represent the change label map and edge label map correspondingly.

IV. Experiments

In this section, in order to validate the effectiveness of the proposed network, we conducted comparison and ablation

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experiments using three publicly available RS CD datasets. These three datasets include the Learning Vision and RS Laboratory building CD (LEVIR-CD) [19] dataset, the Wuhan University (WHU-CD) [5] dataset, and the Sun Yat-Sen University CD (SYSU-CD) [15] dataset. The three CD datasets are described in detail first. Following that, we provide introductions to the compared methods and implementation details used in the experiments. Finally, we present reports and analyses of the comparison experiments conducted with state-of-the-art (SOTA) methods, as well as the ablation studies performed to verify the value of each component in the proposed network.

A. Description of Datasets

1) LEVIR-CD: This dataset is a large-scale very high-resolution (VHR 0.5m/pixel) optical dataset for CD. It comprises 637 pairs of Google Earth images, each with a size of 1024 × 1024 pixels. These VHR images cover 20 different areas situated in various cities in Texas, USA including 31 333 individual change buildings, such as villas, garages, warehouses, and apartments. In the training phase, these original images are cropped to 256 × 256 size for better performance on GPUs. Then, under its official dataset split setting, i.e., 7:1:2, we obtain a total of 7120/1024/2048 image pairs for training/validation/test, respectively.

2) WHU-CD: This public optical dataset contains one pair of HR images with a spatial size of 32 507 × 15 354 and a resolution of 0.075 m captured from Christchurch, New Zealand, that is struck by a devastating 6.3-magnitude earthquake in 2011. Given that the dataset partition resolution is not provided in [5], we conduct a nonoverlapping cropping operation on the original image pairs to acquire small patches with a size of 256 × 256, thereby increasing the volume of available data. In order to conduct fair experimental comparisons, we follow the allocation ratio in BIT [21] and partition the cropped images through a random sampling operation on these patches, which results in the numbers of 6095, 762, and 763, respectively.

3) SYSU-CD: This dataset includes 20 000 pairs of aerial images collected in Hong Kong with 256 × 256 spatial size and 0.5 m spatial resolution, which focus on change information on multifarious types of objects, such as new urban buildings, vegetation changes, road sprawls, and marine constructions. Since most of the current experiments conducted on this dataset in SOTA methods utilize the default allocation ratio of (6/2/2) for sampling, for the sake of fairness, we divide the samples according to the official ratio resulting in 120 000/4000/4000 image pairs for training/validation, respectively.

B. Experimental Setup

1) Implementation Details: We implement the proposed network and all of the other comparison methods with the PyTorch framework. The training and testing processes are carried out on an NVIDIA GeForce RTX3060 GPU. The optimizer chosen for updating the model parameters is stochastic gradient descent (SGD) with a momentum of 0.9 and weight decay of 5e-4. During the training phase, a batch size of 4 and an initial learning rate of 0.01 were set. The learning rate is decayed by the multiplication of 0.3, 0.6, and 0.5 every 25 epochs (100 epochs in total) for LEVIR-CD, WHU-CD, and SYSU-CD datasets, respectively. The tradeoff weight λ for the auxiliary loss is set to 0.25.

2) Evaluation Metrics: Five popular metrics calculated in (20)–(24) are applied in the experiments to measure the performance of the proposed methods, including precision (Pre), recall (Rec), overall accuracy (OA), $F1$-score ($F1$), and intersection over union (IoU). TP and FP represent the numbers of changed pixels that are correctly and wrongly identified, respectively. TN and FN represent the numbers of unchanged pixels that are correctly and wrongly identified, respectively. For each metric, a higher value indicates better performance and the maximum value is 1

$$\text{Pre} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{20}$$

$$\text{Rec} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{21}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{22}$$

$$F1 = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}} \tag{23}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}. \tag{24}$$

C. Comparison With SOTA Methods

To comprehensively showcase the superiority of the proposed network, we have selected several SOTA CD models that are based on DL technology as comparison methods including FC-EF [31], FC-Siam-diff [31], FC-Siam-conc [31], CDNet [27], EGRCNN [41], BIT [21], SNUNet [42], DMINet [43], and P2V-CD [44], which are briefly described as follows.

1) FC-EF [31]: This network adopts a U-Net structure, in which the bi-temporal inputs are concatenated directly and fed into a single-branch network to generate the change map.

2) FC-Siam-diff [31]: It is a modified network based on FC-EF, which introduces a dual-branch encoder for individual input instead of directly disposing of the composite image. The bitemporal features extracted from the dual-branch encoder are conducted by a difference operation to obtain the difference information.

3) FC-Siam-conc [31]: Similar to the dual-branch structure of FC-Siam-diff, this network inputs the image pairs separately and displaces the difference operation with concatenation operation to fuse the feature pairs to obtain change information.

4) CDNet [27]: It is a classic CD network based on FCN architecture, which is used for street scenes. Two components, stacking contraction and expansion blocks, are introduced in the network for pixel-level CD.

5) EGRCNN [41]: This network leverages the difference information and edge prior information simultaneously for better detection results while introducing the LSTM [45] to enhance the change information representation.

6) BIT [21]: This model introduces the transformer [46] based on self-attention to the CD field, with the role of

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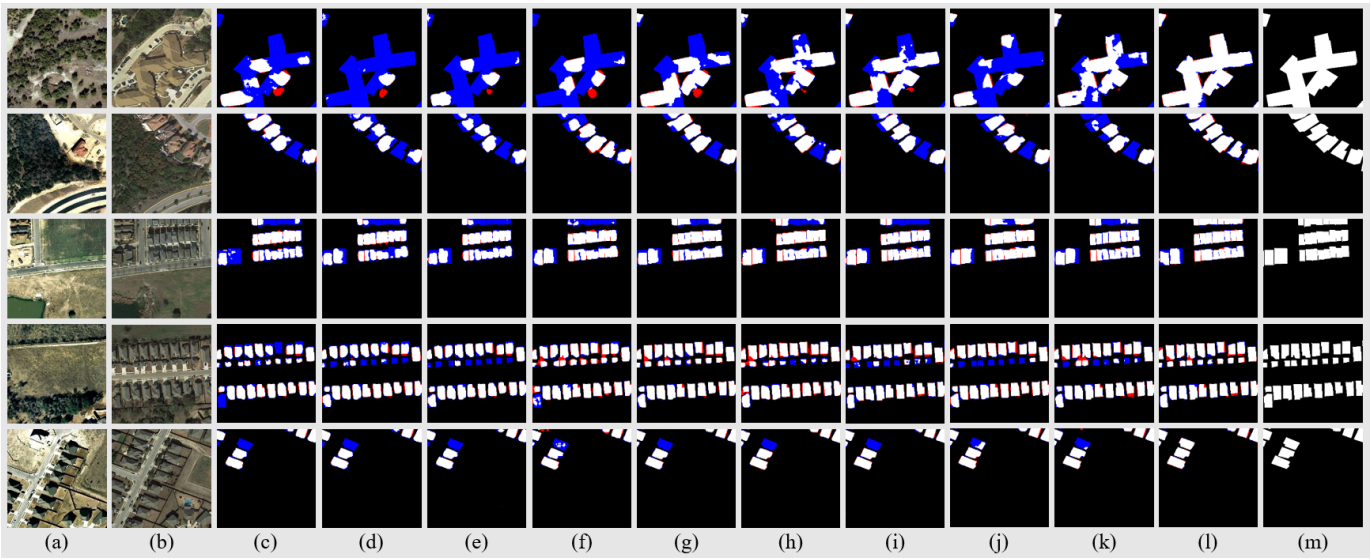


Fig. 6. Qualitative results of the overall methods on the LEVIR-CD dataset. (a) Image T1. (b) Image T2. (c) FC-EF. (d) FC-Siam-diff. (e) FC-Siam-conc. (f) CDNet. (g) EGRCNN. (h) BIT. (i) SNUNet. (j) DMINet. (k) P2V-CD. (l) Ours. (m) Ground truth.

modeling contextual information in spatial–temporal domains and guiding the original features by means of the refined semantic tokens.

7) SNUNet [42]: This network adopts the UNet++ [47] structure, in which a dense skip connection strategy is applied to integrate multiscale features for narrowing the semantic gap. Meanwhile, the CAM [17] is introduced to refine the multilevel semantic features for better performance.

8) DMINet [43]: This network devises a cross-temporal self-attention module for bi-temporal inputs to bond their individual representations and boost their information interaction.

9) P2V-CD [44]: The CD task is viewed as a video understanding problem for the first time in this network, in which dual-branch decoupled encoders composed of spatial branch and temporal branch are introduced to alleviate the problem of incomplete modeling in the temporal domain.

D. Comparing Experiments

In this section, we record the quantitative results of the above methods on the three datasets in Tables I–III and exhibit the qualitative results in Figs. 6–8. Specifically, the highest and the second highest index in the table are highlighted in red and green, respectively. Moreover, the TP, TN, FP, and FN areas in the visual results are marked in white, black, red, and blue correspondingly.

1) Comparative Experiments on LEVIR-CD Dataset: The comparison results of each method on the LEVIR-CD dataset are presented in Table I. It is evident that our proposed method outperforms other comparison methods in most performance indicators, achieving the highest scores OA at 99.18%, $F1$ at 91.53%, and IoU at 84.39%, with improvements of 0.73% and 1.21% in $F1$ and IoU, respectively, compared to the second best method (SNUNet). Concretely, both the FC-Siam-diff and FC-Siam-conc achieve a better performance than FC-EF, showing the superiority of the Siamese-based network in CD tasks. However, such simplistic FCN-based methods, including the typical CDNet, may struggle to adequately extract change information in complex scenes, resulting in unsatisfactory performance. EGRCNN leverages edge prior information to enhance change prediction and achieves acceptable metrics with an $F1$ score of 90.70% and an IoU of 82.98%. Drawn lessons from this idea, we introduce an edge decoder for edge information perception to assist CD. In the context of CD tasks, BIT introduces the use of transformers for the first time. While the performance of BIT may not be particularly impressive in terms of metrics, it still makes a significant contribution to the field of CD by promoting the application and variation of transformers. The SNUNet based on a densely skip

TABLE I
COMPARATIVE EXPERIMENT RESULTS FOR EACH COMPARISON METHOD ON LEVIR-CD DATASET

Methods	Pre(%)	Rec(%)	OA(%)	F1(%)	IoU(%)
FC-EF [31]	88.80	82.11	98.56	85.33	74.41
FC-Siam-diff [31]	93.29	79.79	98.68	86.01	75.45
FC-Siam-conc [31]	**94.10**	79.84	98.72	86.38	76.03
CDNet [27]	90.89	84.47	98.78	87.56	77.88
EGRCNN [41]	93.25	88.28	99.08	90.70	82.98
BIT [21]	91.42	87.74	98.96	89.54	81.06
SNUNet [42]	92.75	**88.97**	**99.08**	**90.82**	**83.18**
DMINet [43]	91.00	88.19	98.95	89.57	81.12
P2V-CD [44]	**93.63**	87.78	99.07	90.61	82.84
Ours	92.87	**90.23**	**99.18**	**91.53**	**84.39**

TABLE II
COMPARATIVE EXPERIMENT RESULTS FOR EACH COMPARISON METHOD ON WHU-CD DATASET

Methods	Pre(%)	Rec(%)	OA(%)	F1(%)	IoU(%)
FC-EF [31]	83.79	79.72	98.48	81.70	69.07
FC-Siam-diff [31]	88.58	79.10	98.67	83.57	71.78
FC-Siam-conc [31]	89.99	82.23	98.85	85.94	75.34
CDNet [27]	92.58	81.68	98.94	86.79	76.66
EGRCNN [41]	89.87	88.68	99.09	89.27	80.62
BIT [21]	92.10	90.30	99.26	91.19	83.80
SNUNet [42]	92.22	**90.49**	99.27	91.34	84.07
DMINet [43]	93.38	88.40	99.24	90.82	83.19
P2V-CD [44]	**96.42**	88.28	**99.36**	**92.17**	**85.48**
Ours	**95.35**	**91.53**	**99.45**	**93.40**	**87.62**

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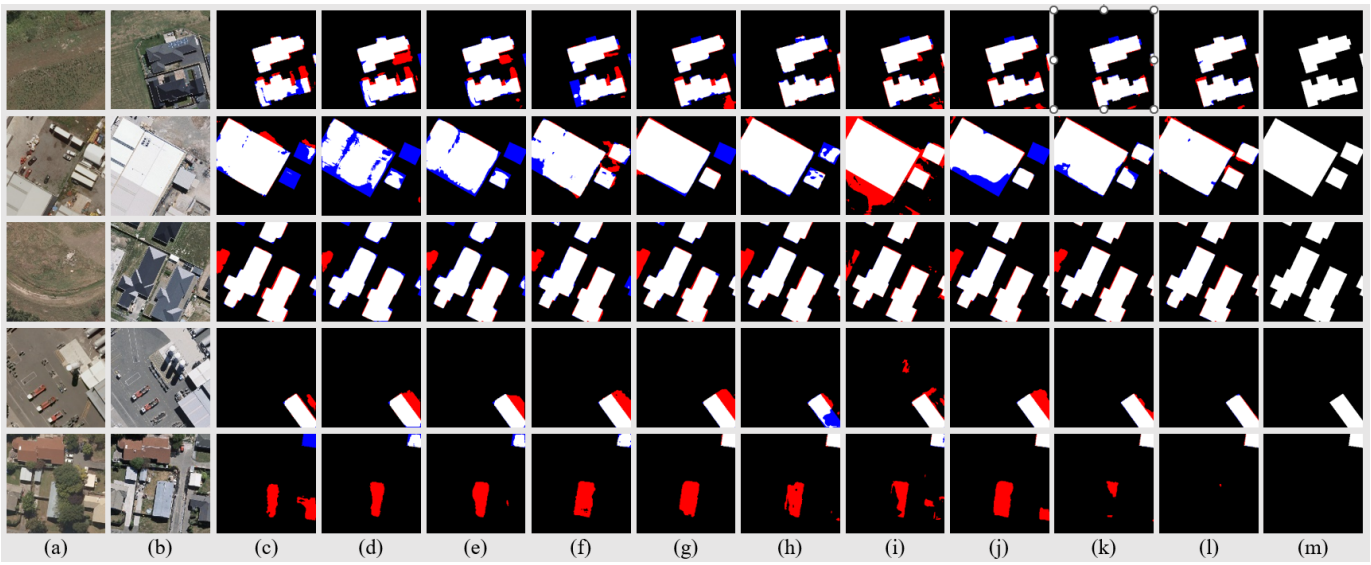


Fig. 7. Qualitative results of the overall methods on the WHU-CD dataset. (a) Image T1. (b) Image T2. (c) FC-EF. (d) FC-Siam-diff. (e) FC-Siam-conc. (f) CDNet. (g) EGRCNN. (h) BIT. (i) SNUNet. (j) DMINet. (k) P2V-CD. (l) Ours. (m) Ground truth.

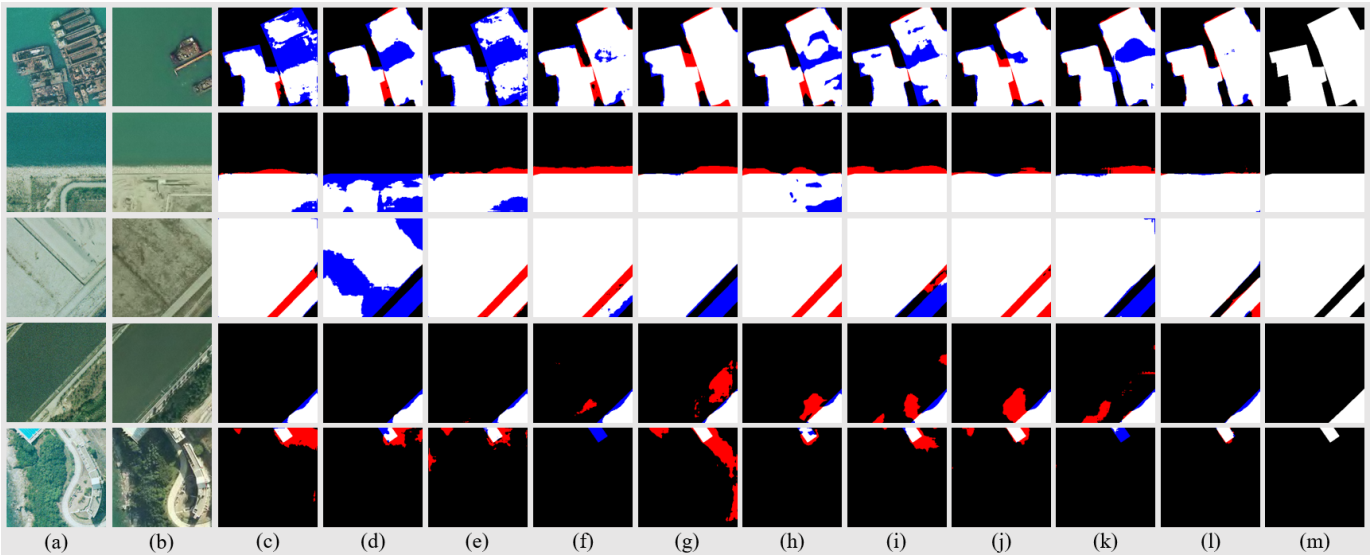


Fig. 8. Qualitative results of the overall methods on the SYSU-CD dataset. (a) Image T1. (b) Image T2. (c) FC-EF. (d) FC-Siam-diff. (e) FC-Siam-conc. (f) CDNet. (g) EGRCNN. (h) BIT. (i) SNUNet. (j) DMINet. (k) P2V-CD. (l) Ours. (m) Ground truth.

TABLE III
COMPARATIVE EXPERIMENT RESULTS FOR EACH COMPARISON METHOD ON SYSU-CD DATASET

Methods	Pre(%)	Rec(%)	OA(%)	F1(%)	IoU(%)
FC-EF [31]	78.07	71.73	88.58	74.77	59.70
FC-Siam-diff [31]	**88.95**	54.90	87.76	67.90	51.40
FC-Siam-conc [31]	79.15	74.27	89.32	76.63	62.12
CDNet [27]	77.19	78.86	89.52	78.01	63.95
EGRCNN [41]	80.77	**81.02**	90.98	**80.90**	**67.92**
BIT [21]	**84.56**	72.37	90.37	77.99	63.92
SNUNet [42]	82.49	79.02	**91.20**	80.89	67.91
DMINet [43]	82.04	79.04	90.97	80.51	67.37
P2V-CD [44]	84.33	75.06	90.83	79.42	65.87
Ours	82.68	**80.51**	**91.43**	**81.58**	**68.89**

connection strategy achieves the second-highest scores on the LEVIR-CD dataset among overall methods, which obtains the $F1$ and IoU of 90.82% and 83.18%, respectively. The cross-temporal self-attention module devised in DMINet boosts the information interaction of bi-temporal images, which achieves the $F1$ and IoU of 89.57% and 81.12%. The final comparison method P2V-CD, with the $F1$ and IoU of 90.61% and 82.84%, regards the CD problem as a video understanding task, which leads to a promising direction for the CD field. By thoroughly evaluating and selectively incorporating a subset of the strategies used in the aforementioned methods, our proposed method surpasses them by a significant margin in terms of performance.

The perceptual visualization results of the various methods on the LEVIR-CD dataset are given in Fig. 6. The LEVIR-CD dataset primarily consists of compact and complex small-changed buildings. These structures pose challenges in CD due to their intricate nature and susceptibility to the interference of external environmental factors. The scene in the first group exists as an irregular and continuously changed area. Among the samples, our method stands out as the only one capable of achieving almost complete detection of the change regions,

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which highlights the robust ability of our method to effectively handle complex targets.

In the second group of samples, the comparison methods exhibit a less precise representation of the changed building details compared to our method. This superior performance can be attributed to the effective mitigation of interclass indistinction and intraclass inconsistency, which is achieved through the novel triple-branch structure of the encoder in our proposed method.

The third group of results can measure the ability to detect densely arranged buildings. As can be seen, the top one of the three lines of compact buildings in the binary maps is seriously missed in the simple FCN-based methods including FC-EF, FC-Siam-diff, FC-Siam-conc, and CDNet. While the problem of missed detections is slightly alleviated in BIT, DMINet, and P2V-CD, the results are still not entirely satisfactory. In stark contrast, our method demonstrates a more impressive performance in detecting changes within the same areas, with almost no missed predictions.

In the fourth group of results, it can be observed that many comparison methods suffer from serious missing of small targets. While CDNet and EGRCNN exhibit fewer instances of this phenomenon, they still generate numerous false positives primarily due to the interference of lighting conditions and shadows appearing on the change regions. Relying on the multiple sizes of receptive field applied in the JFC module, our method can effectively capture multiscale objects, encompassing even small changed targets.

The last group of results focuses on comparing the ability to withstand the interference of lighting. In this evaluation, all of the comparison methods fail to pass the test of anti-interference, as they misclassify a block of buildings affected by the lighting factor. However, in the same case, our method correctly identifies the affected changed regions. The above five groups of qualitative results on LEVIR-CD powerfully validate the robustness and effectiveness of our method.

2) Comparative Experiments on WHU-CD Dataset: The performance metrics of the comparative methods tested on the WHU-CD dataset are displayed in Table II. It is evident that our method achieves the highest scores in most indicators such as Rec, OA, $F1$, and IoU. The $F1$ and IoU of our method are 91.53% and 84.39%, with improvements of 1.23% and 2.14%, respectively, compared to the second-best one P2V-CD. Due to the large number of large-scale building targets in WHU-CD, the contribution of leveraging edge information in EGRCNN is relatively not significant, resulting in poor scores on $F1$ and IoU. Relying on the ability to model global contextual relationships, both BIT and DMINet achieve acceptable but not impressive results on $F1$ and IoU. Benefiting from the incorporation of densely connected skip connection operations and the CA mechanism, the SNUNet achieves a satisfactory result in $F1$ and IoU of 91.34% and 84.07%, respectively. Second only to our method, the P2V-CD achieves scores of 92.17% and 85.48% in $F1$ and IoU, respectively.

Fig. 7 shows five groups of representative visualization samples of each method on the WHU-CD dataset. Similar to the LEVIR-CD dataset, the WHU-CD dataset mainly concentrates on building changes. However, the scale of building targets in WHU-CD is generally larger than those in LEVIR-CD, which poses a specific challenge for CD methods to strive to achieve complete identification of continuous and large-scale changed regions. Specifically, the first group contains two large-scale buildings with complex structures, in which the irregular contour structure leaves an arduous challenge for the network. Due to the similarity in color between the roof and the ground, four FCN-based typical methods erroneously identify numerous negative regions as the changed targets. Though these similar incorrect recognitions have been reduced in the remaining comparison methods, the processing of local detail information is unsatisfactory, resulting in some false negatives. Facing the same challenge, almost no true positives are missed in our method with only a small range of false predictions.

The changed targets in the second group contain large-scale and two small-scale regular buildings, which test the capacity to perceive multiscale objects and maintain the integrity and continuity of the large-scale target regions. Both two standards have not been well completed in FC-EF, FC-Siam-diff, FC-Saim-conc, and DMINet. As for CDNet, the integrity prediction of the large-changed building is unacceptable while there is a lack of effective discrimination against noise around the small-scale targets. EGRCNN and BIT demonstrate impressive performance in large-scale object detection, yet fail to achieve equal performance when it comes to small-scale object recognition. SNUNet achieves a brilliant recognition of both large-scale and small-scale targets yet suffers the most from the interference of noise, with a large amount of false positive areas. P2V-CD, as the second best one, achieves a significant improvement in both standards compared to previous methods but is still a bit inferior compared to our method.

In the third group, all comparison methods incorrectly regard a piece of ground as a changed building except for our method. The last two groups of visualization results display the impact of lighting and external obstruction on CD, respectively. Specifically, for the comparison methods in the fourth group, the discrimination of the changed building is seriously affected by the different lighting conditions. In the fifth group, as the changed building is covered by dense vegetation, the comparison methods cannot correctly pick out the true positives yet our method can. The extraordinary performances in both disturbed situations strongly prove the effectiveness and robustness of our network for CD tasks.

3) Comparative Experiments on SYSU-CD Dataset: Table III reports the test results of each comparison method on the SYSU-CD dataset. Since the diverse type of change information that the SYSU-CD focused on encompasses not only building targets, the performance of evaluation metrics on each method is not as good as the performance on the previous two datasets. Even so, our method is still superior to the other SOTA methods. Specifically, the best performances of $F1$ and IoU attained by our method are achieved at 81.58% and 68.89%, which are 0.68% and 0.97% higher than the second-highest scores obtained by EGRCNN. The performances of the three classic FCN-based methods are more inconspicuous in such complicated scenes. The CDNet and BIT achieve the approximate behaviors with the $F1$ and IoU of 78.01% and 63.95%, 77.99%, and 63.92%, correspond-

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TABLE IV
EVALUATION OF NETWORK WITH DIFFERENT ARCHITECTURES ON THREE DATASETS

Architecture	Single-enc	Dual-enc	Triple-enc	Sing-dec	Double-dec	LEVIR-CD			WHU-CD			SYSU-CD		
						Pre(%)	Rec(%)	F1(%)	Pre(%)	Rec(%)	F1(%)	Pre(%)	Rec(%)	F1(%)
Structure1		✓		✓		**93.60**	84.32	88.72	86.61	90.90	88.70	81.46	75.88	78.57
Structure2	✓			✓		91.25	87.80	89.49	92.31	86.43	89.27	**90.60**	77.13	78.82
Structure3			✓	✓		92.65	87.88	90.21	91.42	91.32	91.37	80.08	81.10	80.59
Structure4	✓				✓	91.81	87.50	89.60	**95.12**	86.59	90.66	76.71	**81.21**	78.90
Structure5			✓		✓	92.55	**88.89**	**90.68**	92.55	**91.32**	**91.93**	81.48	80.33	**80.90**

TABLE V
EVALUATION OF VARIOUS DESIGNED MODULES ON THREE DATASETS

Variants	MDG	CCA	LEVIR-CD			WHU-CD			SYSU-CD		
			Pre(%)	Rec(%)	F1(%)	Pre(%)	Rec(%)	F1(%)	Pre(%)	(%)	F1(%)
Structure1 +			**93.60**	84.32	88.72	89.72	85.57	87.60	81.46	75.88	78.57
Structure1 +	✓		**93.69**	87.06	90.25	**89.93**	90.30	90.11	80.92	**78.78**	79.84
Structure1 +		✓	92.50	88.64	90.53	89.55	91.30	90.42	**83.79**	76.50	79.98
Structure1 +	✓	✓	91.47	**89.83**	**90.64**	88.75	**92.67**	**90.67**	82.69	78.12	**80.34**

ingly. The indicators obtained by SNUNet approach to the second highest ones, with the $F1$ and IoU at 80.89% and 67.91%, respectively. The two recent methods, DMINet and P2V-CD, perform uneventfully on SYSU-CD with $F1$ and IoU of 80.51% and 67.37%, 79.42% and 65.87%, correspondingly.

To exhibit the performance of each method in diverse change situations, five groups of visualization results of various changes are shown in Fig. 8. For the change situation of sea construction in the first group of results, massive positive omission issues occur in the three classic FCN-based methods, SNUNet and P2V-CD while existing some misclassified regions around the changed areas in the other comparation methods. In contrast, our method performs more effectively in alleviating these two types of issues. The next two groups of samples contain the change situation of large-scale groundwork before construction, which are suitable to test the ability to detect the boundaries between changed and unchanged areas.

For the detection of the small-scale road expansion in the fourth group, the relatively complex methods yield different scopes of false predictions beyond the correct areas. By contrast, the three classic FCN-based methods with simple structures obtain excellent predictions in this situation. It means that the quality of prediction results is not determined by the complexity of the model. Significantly, our method outperforms the three FCN-based methods with a comparable complexity to the previously mentioned complex approaches.

E. Ablation Experiments

To evaluate the contribution of each proposed component of our method, we conduct ablation experiments on the three abovementioned datasets, in which the implementation details in all participating networks are set to the same. Specifically, the ablation experiments mainly revolve around two parts: different network architectures and various designed modules, the results of each part are shown in Tables IV and V, respectively.

1) Effectiveness of the TD Framework: The TD framework adopted in our network is composed of a triple-branch encoder and a double-branch decoder. To evaluate the efficacy of different architectures of the network, we implement five ablation networks based on different combinations of various encoder and decoder forms. As shown in Table IV, Single-enc and Dual-enc denote the individual utilization of the two streams within the Triple-branch encoder, respectively. Single-dec indicates the exclusive use of the change decoder. First, we conduct the top three ablation studies by combining three different encoder configurations with Single-dec to confirm the effectiveness of simultaneously learning both the individual and joint features of bi-temporal images. Attributed to the adaptive multiscale learning capability of JFC modules in the single-branch encoder, Structure2 exhibits slightly better performance compared to Structure1 across all three datasets. By combining the strengths and mitigating the weaknesses of the dual-branch and single-branch encoders, Structure3 achieves great progress compared to the previous two variants, with $F1$ increasing by 1.49% and 0.72%, 2.67% and 2.10%, and 2.02% and 1.77% on LEVIR-CD, WHU-CD, and SYSU-CD datasets, respectively. The results strongly indicate the superiority of the triple-branch encoder.

To assess the impact of the edge auxiliary decoder, we conduct two ablation studies by pairing the single-branch encoder and triple-branch encoder with the double-branch decoder (Double-dec), respectively. Obviously, with the guidance of the detailed edge information, both two variants have achieved notable progress on three datasets. When compared to their counterparts that use the corresponding encoder but independently employ a change decoder, both Structure4 and Structure5 make significant advancements on the first two datasets. However, the improvement on the SYSU-CD dataset is not impressive due to its more complex changed scenarios. Accordingly, datasets featuring sharper-edged changed regions, such as buildings, derive greater benefits from the implementation of supervision of the edge structure informa-

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tion. The comprehensive analysis of the results of two ablation experiments on network architecture indicates the validity and effectiveness of the proposed TD framework.

2) Effectiveness of the MDG and CCA Modules: As shown in Table V, to validate the efficacy of the designed modules in our method, we select Structure1 as the base network. We evaluate the effectiveness of each proposed module by incrementally incorporating the MDG module, CCA module, and their combined application into the base network. Both two modules are applied to the dual-encoder phase to facilitate information interaction between the bi-temporal features.

a) MDG module: The MDG module leverages the enhanced difference information obtained by the multiscale pooling operation and channel-attention guidance to recalibrate the bi-temporal features. To specifically validate the effectiveness of the MDG module, we remove the subsequent CCA module and directly conduct subtraction operations between out_1 and out_2 in Fig. 2. The subtracted features are treated as the output of the encoder at the corresponding stage, while the out_1 and out_2 are transmitted to the next encoder stage. Compared to the base network, the separate additional application of MDG in the encoder produces considerable progress in performance, with the $F1$ increasing 1.53%, 2.51%, and 1.27% on the LEVIR-CD, WHU-CD, and SYSU-CD datasets, respectively. The positive impact of MDG underscores the importance of extracting local and global difference features, which contribute significantly to refining the bi-temporal features through mutual information interaction.

b) CCA module: The CCA module, based on a cross-self-attention mechanism, is designed to further promote the interaction between the bi-temporal features. Similarly, to assess its contribution to the network, we remove the MDG module and directly feed the features from the dual-branch encoder into the CCA module to model the mutual global context relationships. The Out_1 and Out_2 in Fig. 4 are treated as the input of the next stage of encoder, while the output of current stage of encoder is the concatenation of Out_1 and Out_2. Compared to the separate application of the MDG module, the introduction of the CCA module for the network yields more outstanding performance. Based on Structure1, the employment of the CCA module increases $F1$ by 1.53%, 2.72%, and 1.41%, respectively. The performance highlights the positive significance of modeling the affinity between mutual information in bi-temporal features.

c) Combination of MDG and CCA: Both the MDG module and CCA module contribute to varying degrees of improvement in network performance. To leverage the difference information guiding ability of the MDG module and the mutual affinity modeling ability of the CCA module simultaneously, we introduce these modules sequentially for features from the dual-branch encoder. Specifically, the bi-temporal features are initially passed through the MDG module for calibration, and then the out_1 and out_2 in Fig. 2 serve as input pairs of CCA module while also being input to the next stage of encoder concurrently. Following mutual interested information perceiving in CCA module, the Out_1 and Out_2 in Fig. 4 are concatenated as the output of current stage. As a result of the combined action of these two modules, the network performance shows a superimposed effect compared to their separate use, with $F1$ increasing by 1.79%, 3.07%, and 1.77% on LEVIR-CD, WHU-CD, and SYSU-CD datasets, respectively, in comparison to the based network. This performance demonstrates the complementary relationship between the MDG module and the CCA module in our network.

TABLE VI
EFFICIENCY EVALUATION OF CCA MODULE ON LEVIR-CD DATASETS

Methods	Pre(%)	Rec(%)	F1(%)
SA-solely	92.76	86.93	89.75
CA-solely	**93.83**	85.60	89.53
SA+CA(Seq)	92.99	87.22	90.01
SA+CA(Par)	93.12	87.23	90.08
CCA	93.08	**87.69**	**90.30**

F. Discussion

1) Discussion on CCA Versus Typical Nonlocal Attention Mechanisms: To validate the efficiency and superiority of the proposed attention mechanism CCA, we conducted a comparative analysis with the SA mechanism, CA mechanism, and their simple combinations: SA+CA(Seq), SA+CA(Par). In order to perform this comparison, we reimplemented four models that incorporate these attention mechanisms. Specifically, we select Structure1 as the base network and implement the four mechanisms and CCA exclusively in the second layer of dual-branch encoders, respectively. To ensure consistency with CCA, all the comparative attention mechanisms adopt a cross-attention manner for affinity modeling of bi-temporal features. For the sake of fairness, we maintain consistent hyper-parameter settings and data partition across all models. The Pre, Rec, and $F1$ on the LEVIR-CD dataset for each method are shown in Table VI. Due to the attention loss and bias, the SA-solely and CA-solely achieve lower scores in $F1$ compared to the models that combined both mechanisms in different manners. However, the integrity and interaction of features refined by two attention mechanisms are implemented as a postprocessing step, resulting in another attention bias. In contrast, the proposed CCA mechanism conjointly models the contextual affinity across channels and spatial within a unique attention matrix. With the guidance of richer inference clues, CCA achieves an impressive improvement with the $F1$ of 90.30%.

Moreover, to demonstrate the superiority of the computational efficiency of the CCA mechanism in terms of both inference time and floating-point operations (FLOPs), we calculated and presented the values for each attention mechanism using a bar chart in Fig. 9. The inference time is calculated as an average time for 300 subpatches for each method. And the FLOPs are calculated with the feature size of $128 \times 128 \times 64$. Obviously, the proposed CCA mechanism demonstrates a similar level of inference time compared to CA. Notably, both CCA and CA mechanisms exhibit significantly lower inference times than SA and the two combined attention mechanisms. As for FLOPs, the proposed CCA mechanism achieves a slightly higher value than the CA mechanism. The FLOPs for

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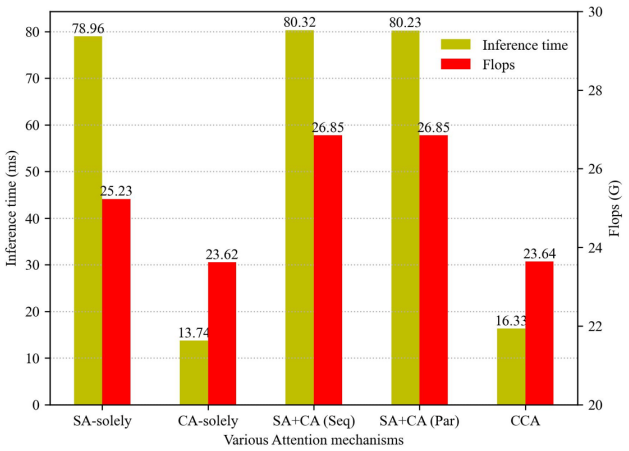


Fig. 9. Effectiveness analysis results of various attention mechanisms based on Structure1. Inference time is calculated as an average time for 300 experiments.

TABLE VII
ABLATION STUDY ON THE POOLING SCALES OF MDG MODULES ON LEVIR-CD DATASETS

Pooling scales			LEVIR-CD		
1	2	4	Pre(%)	Rec(%)	F1(%)
✓			91.04	**88.87**	89.94
	✓		91.98	88.11	90.00
		✓	92.47	87.54	89.94
✓	✓		93.05	87.37	90.12
	✓	✓	92.20	88.03	90.07
✓	✓	✓	**93.69**	87.06	**90.25**

SA are much higher than the CA and CCA mechanism yet lower than SA + CA (Seq) and SA + CA (Parallel).

In conclusion, the proposed CCA mechanism offers several advantages over typical nonlocal attention mechanisms. It allows the network to effectively model contextual affinity by leveraging richer clues, resulting in improved performance in tasks such as feature refinement and interaction while maintaining a promising inference speed and low requirement for FLOPs. These factors highlight the practicality and efficiency of the CCA mechanism for various applications requiring attention-based models.

2) Discussion on the Pooling Scales of the MDG Modules: The MDG module leverages the different information at different scales to guide the concentration of changed areas in bi-temporal features during the encoder phase. Therefore, selecting the appropriate parallel pooling scales is crucial for optimal performance. To discuss the selection of pooling scales in the MDG module, we chose Structure1 as the base network and experimented with several variants of the MDG module by introducing various pooling scales. According to the results presented in Table VII, it can be observed that using a single pooling scale from the candidate options in the top three rows yielded unremarkable performance, with approaching accuracy levels but not surpassing expectations. However, when employing two pooling operations in parallel with adjacent scales, a slight improvement in performance was achieved. Notably, the combination of pooling operations at scales 1 and 2 slightly outperforms the combination of scales 2 and 4, since the former combination enables the network to incorporate both local and global context. Following this observation, we parallelly apply three pooling operations with all candidate scales. With more diverse features generated from multiple scales, the MDG module can attain a more comprehensive understanding of the different information within the bi-temporal data, leading to the best performance with the $F1$ of 90.25%. Based on the above discussion, the pooling scales with (1, 2, 4) are selected for the proposed MDG module for generating the enhanced difference information.

TABLE VIII
ABLATION STUDY ON THE CALIBRATION OF MULTISCALE FEATURES IN JFC MODULE ON LEVIR-CD DATASETS

Dilation rates				Calibration	LEVIR-CD		
1	2	4	6		Pre(%)	Rec(%)	F1(%)
✓					**92.80**	86.01	89.28
	✓				92.68	85.93	89.18
		✓			91.47	86.83	89.09
			✓		91.48	86.66	89.00
✓	✓	✓	✓		92.38	86.42	89.30
✓	✓	✓	✓	✓	91.25	**87.80**	**89.49**

3) Discussion on the Efficacy of the Calibration of Multiscale Features in JFC Modules: The JFC module captures features from multisize fields and calibrates them to guide the network in learning discriminative joint features at multiple scales. To validate the effectiveness of the calibration vector, which encompasses diverse features from multiple scales, we conducted five ablation studies by using convolution layers with various implements of dilation rates. Specifically, for the first four ablation studies, we replace the four parallel multiscale convolution operations and the subsequent calibration process between the two 1×1 convolution layer in Fig. 4 with a single convolution layer of a specific dilation rate. In addition, as the fifth ablation experiment, we reserve the four parallel convolution layers in the JFC module while removing the calibration operations to validate the significance of the calibration vector. As shown in Table VIII, the first four variants demonstrate a decreasing trend in terms of $F1$ with increasing dilation rates. The reason for this trend is that the sparse points sampled by a large dilation rate cannot adequately capture the small targets and detailed information. However, convolutions with larger dilation rates can capture broader global information. Therefore, the fifth ablation experiment involves the parallel implementation of four convolution layers with different dilation rates to simultaneously capture both local information and global information, resulting in a slight improvement over the convolution with a dilation rate of 1 in terms of $F1$. Considering that features at different scales may exhibit varying importance in different scenarios, we additionally introduce a calibration operation for the multiscale features, enabling the adaptive assignment of different weights to them based on their significance. With the guidance of the calibration vector in the JFC module, the performance achieves the highest $F1$ score of 89.49%.

V. CONCLUSION

In this article, a TD framework with a CCA modeling network is proposed for CD tasks on RS images, where a

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Triple-branch encoder consists of both two types of streams to complement their advantages and disadvantages. In a dual-branch encoder, the MDG module leverages enhanced difference information at multiple scales to guide the original features. Following this guidance, the calibrated feature pairs are fed to the CCA module for further information interaction through the modified self-attention mechanism to model channel dependencies and spatial relationships simultaneously. In addition, for more sufficient extraction of joint features, we design the JFC module to construct the single-branch encoder, in which the atrous convolutions are employed to perceive features at various scales and use the CA mechanism to reweight these diverse feature maps. In the phase of image reconstruction, in contrast to the single-branch decoder-based UNet structure, we introduce a double-branch decoder for feature fusion and edge information optimization, which achieves better performance on change prediction. Experimental results indicate that the proposed outperforms other SOTA CD methods.

Considering that, as a fully-supervised method, the proposed network heavily relies on pixel-level labels, which are labor-intensive and time-consuming to obtain. In addition, the limitation of pixel-wise annotations, which may not always be available or feasible for large-scale RS datasets, restricts the scalability and applicability of fully supervised methods in some cases. Therefore, to reduce the dependency on fully annotated datasets, we may focus our work on unsupervised or weakly-supervised methods in the future.

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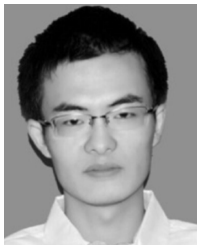
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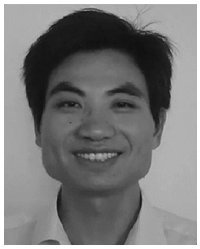
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